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(54) **WIRELESS COMMUNICATION APPARATUS**

(56)

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H01P 1/26 (2006.01)

H01P 5/00 (2006.01)

(52) **U.S. Cl.**

CPC . **H01P 5/00** (2013.01); **H01P 1/26** (2013.01)

(58) **Field of Classification Search**

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See application file for complete search history.

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ABSTRACT

In a wireless communication apparatus including a communication coupler and a circuit, a first substrate and a second substrate face each other in a longitudinal direction of the communication coupler. The first substrate includes a first signal line, a second signal line, a third signal line, a distributor, and a terminating resistor. At the distributor, the first signal line branches into the second signal line and the third signal line. The second signal line is terminated at the terminating resistor at an end of the first substrate. The second substrate includes a fourth signal line. At least one of the third signal line or the connection signal line is configured such that a phase of a signal at the facing portion of the second signal line and a phase of a signal at the facing portion of the fourth signal line are continuous.

3 Claims, 4 Drawing Sheets

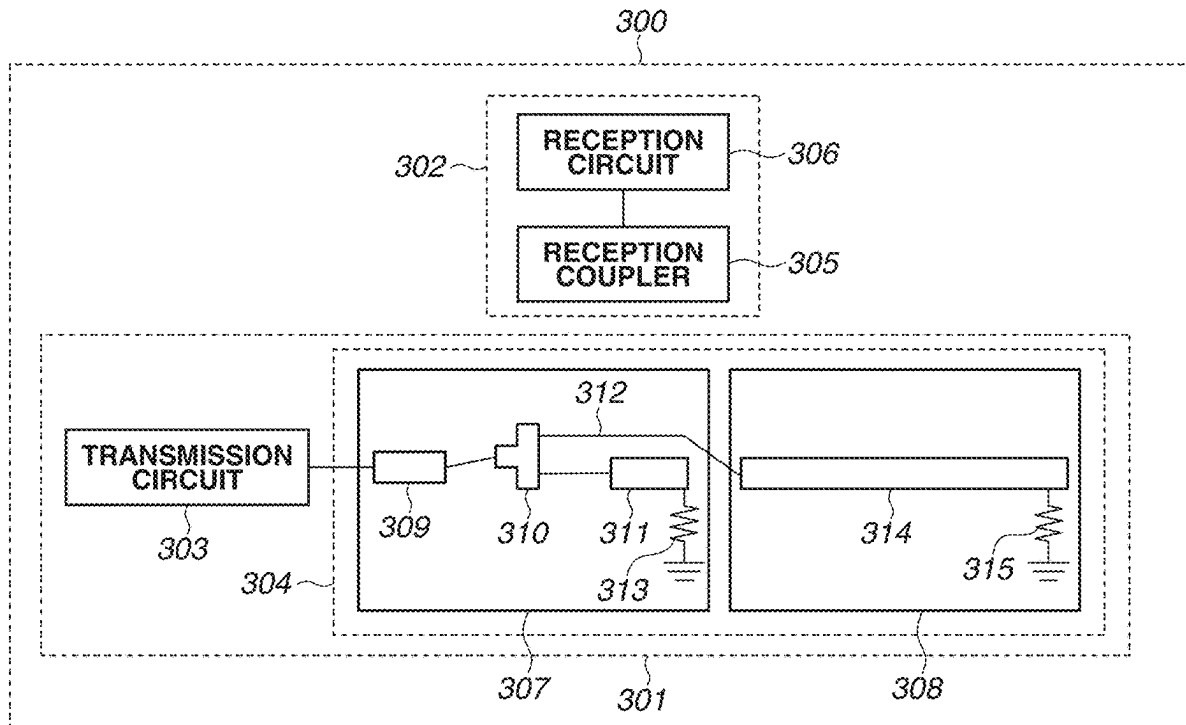


FIG.1A

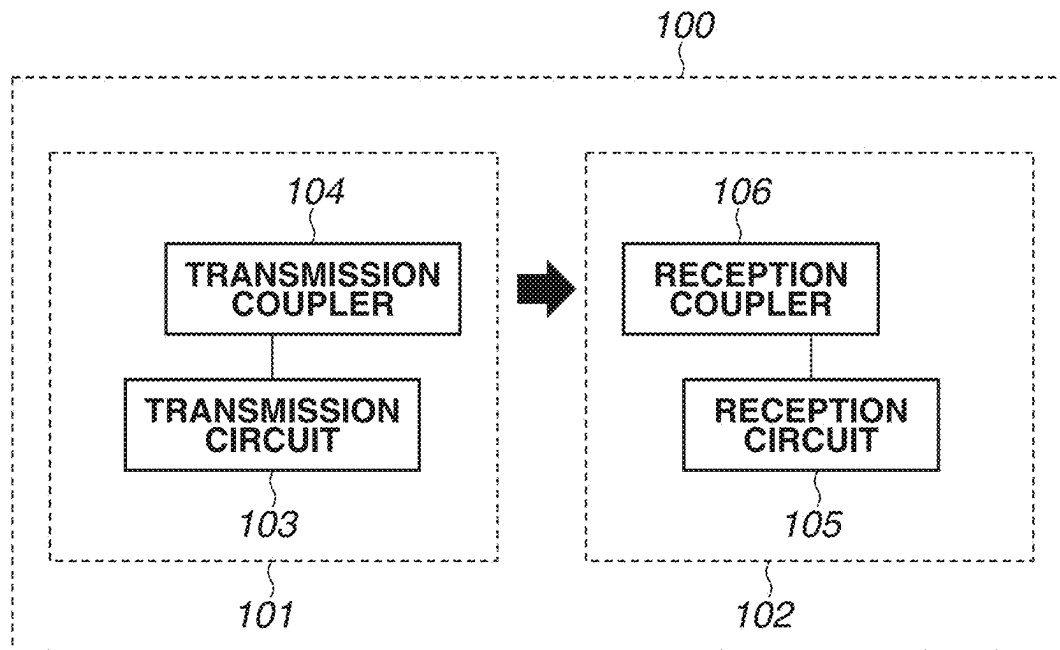


FIG.1B

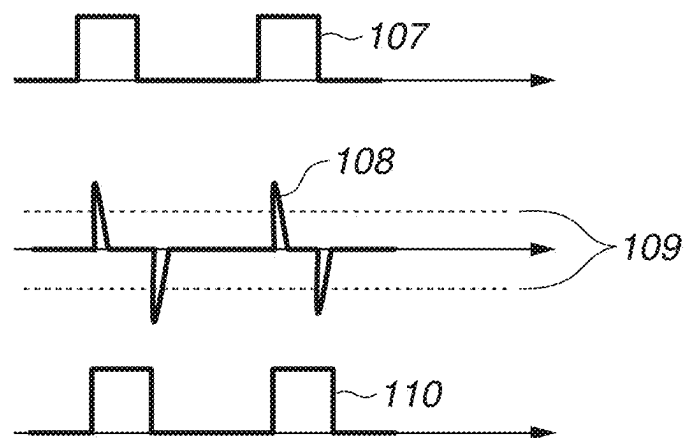


FIG.2A

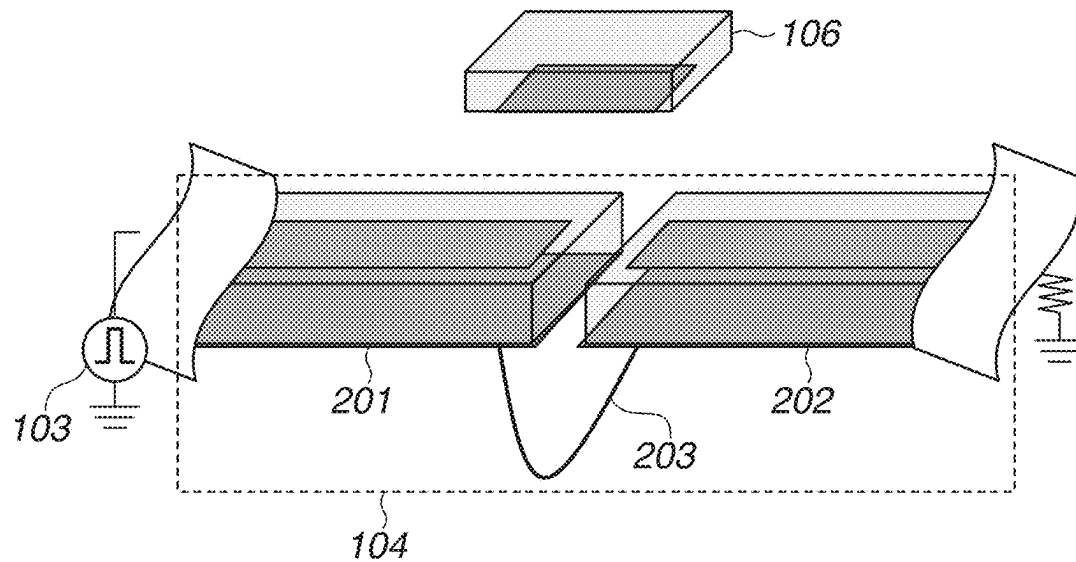


FIG.2B

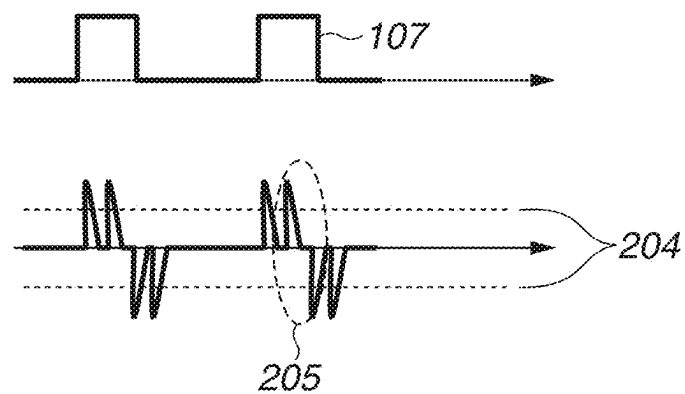


FIG. 3

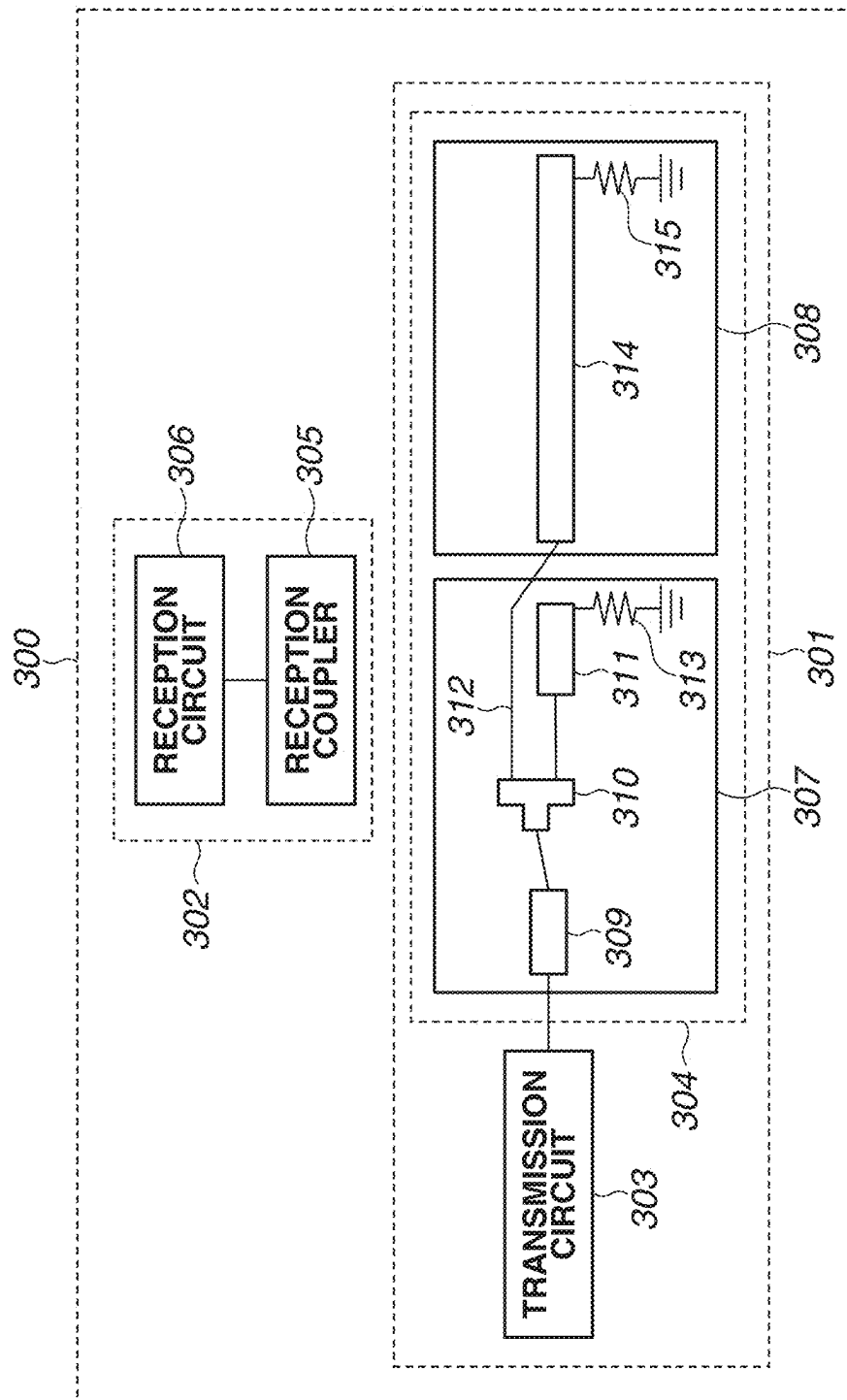
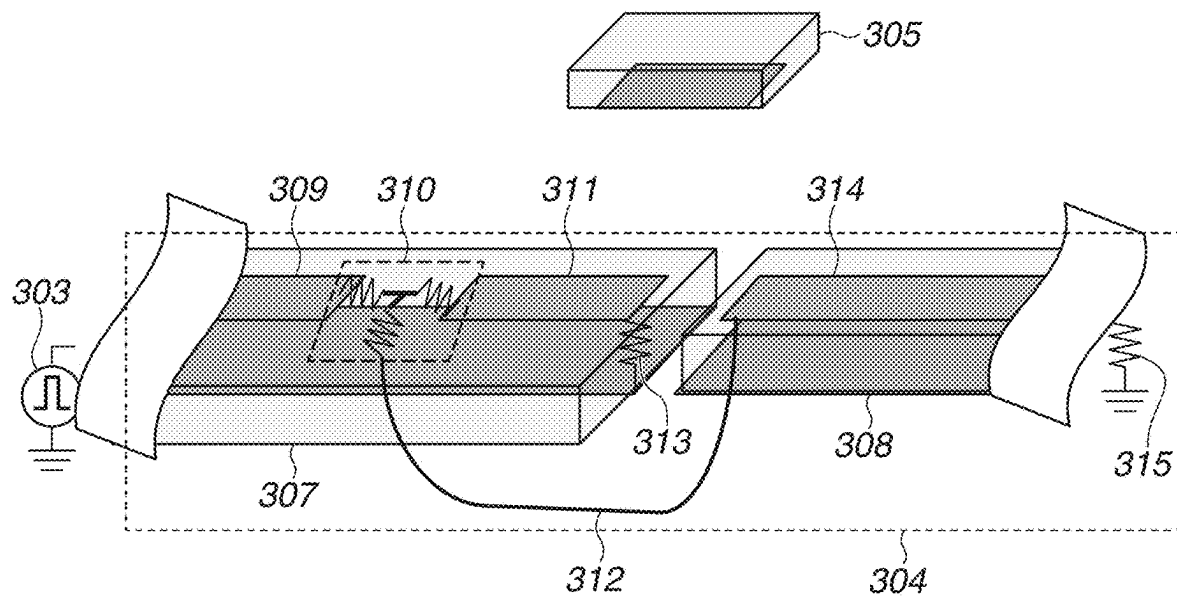


FIG. 4



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WIRELESS COMMUNICATION APPARATUS**BACKGROUND****Field of the Disclosure**

The present disclosure relates to wireless communication techniques.

Description of the Related Art

Wireless communication systems that perform communications by electromagnetic coupling have been developed in recent years. Such a wireless communication system communicates non-modulated baseband signals, which provides high-speed, less-delayed communications with simplified circuitry. For example, Japanese Patent Application Laid-Open No. 2009-268022 discusses a technique of arranging two slot antennas opposed to each other in close proximity, and transferring baseband signals in a non-contact manner.

In wireless communication systems that perform communications between communication couplers disposed on their respective substrates, the length of each communication coupler is restricted to the size of the corresponding substrate. To solve this issue, by connecting a plurality of substrates with a coaxial cable or the like, a communication coupler can be configured without being restricted to the size of a substrate. However, the cable connecting these substrates can cause a signal to be delayed, causing splits at rising and falling edges of a reception signal waveform.

SUMMARY

Aspects of the present disclosure are directed to providing high-speed wireless communications by reducing splits at rising and falling edges of reception signal waveforms.

According to an aspect of the present disclosure, a wireless communication apparatus includes a communication coupler including a plurality of substrates including a first substrate and a second substrate, and a connection signal line, and a circuit configured to transmit or receive a signal to or from the communication coupler. The first substrate and the second substrate are opposed to each other in a longitudinal direction of the communication coupler. The first substrate includes a first signal line, a second signal line, a third signal line, a distributor, and a terminating resistor. At the distributor, the first signal line branches into the second signal line and the third signal line. The second signal line is terminated at the terminating resistor at an end of the first substrate. The second substrate includes a fourth signal line. The connection signal line connects the third signal line and the fourth signal line. A facing portion of the second signal line and a facing portion of the fourth signal line are opposed to each other, and at least one of the third signal line or the connection signal line is configured such that a phase of a signal at the facing portion of the second signal line and a phase of a signal at the facing portion of the fourth signal line are continuous.

Further features of the present disclosure will become apparent from the following description of embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A illustrates the configuration of a wireless communication system according to conventional art, and FIG. 1B illustrates examples of signals that are transmitted and received.

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FIG. 2A illustrates the configuration that lengthens a communication coupler, and FIG. 2B illustrates signal waveforms, according to the conventional art.

FIG. 3 illustrates a wireless communication system according to an embodiment.

FIG. 4 illustrates the configuration of a communication coupler according to the present embodiment.

DESCRIPTION OF THE EMBODIMENTS

Hereinafter, an embodiment will be described with reference to the accompanying drawings. The embodiment described below do not limit the present disclosure. Not all combinations of features described in the embodiment are used as means for solution in the present disclosure.

Issues with the conventional wireless communication technology will be described first. A wireless communication system **100** illustrated in FIG. 1A includes a transmission device **101** and a reception device **102**. The transmission device **101** includes a transmission circuit **103** and a transmission coupler **104**. The reception device **102** includes a reception circuit **105** and a reception coupler **106**. FIG. 1B illustrates an example of signals that are transmitted and received. The transmission coupler **104** in the transmission device **101** transmits a digital signal **107** without modulation, and the reception coupler **106** in the reception device **102** receives a reception signal **108** as a differentiated signal. The reception device **102** restores the reception signal **108** to a signal **110** based on the reception signal **108** and thresholds **109** of a comparator in the reception circuit **105**.

Suppose that communication is being performed while a short sized coupler is being moved in the longitudinal direction of a long sized coupler in the wireless communication system that includes the short sized coupler and the long sized coupler. When the long sized coupler is configured on a substrate, the length of the long sized coupler is restricted to the size of the substrate. However, the long sized coupler can be configured on a plurality of substrates that are connected to each other with a coaxial cable (connection signal line) or the like, without being restricted to the size of a substrate. FIG. 2A illustrates a configuration example in a case of extending the length of the long sized coupler using a coaxial cable. A substrate **201** and a substrate **202** are connected to each other with a coaxial cable **203**. Such a configuration of extending the length of the long sized coupler with the coaxial cable **203** can delay a signal due to influence of the coaxial cable **203**. This signal delay causes splits illustrated in FIG. 2B at rising and falling edges of a reception signal waveform. These splits at the rising and falling edges of the reception signal waveform can reduce the amplitudes at the rising and falling edges, reaching short of thresholds **204** of the comparator. In addition, an interval **205** between a rising edge and the next falling edge will shorten at higher communication speeds. This will cause the amplitudes at the next rising and falling edges to decrease, creating a communication error.

A wireless communication system that can reduce splits at rising and falling edges of a reception signal waveform even with an extended length of the long sized coupler, according to the present embodiment, will now be described.

FIG. 3 illustrates a configuration of the wireless communication system **300** according to the present embodiment. A wireless communication system **300** includes a wireless communication apparatus **301** (transmission device) and a wireless communication apparatus **302** (reception device). The wireless communication apparatus **301** includes a transmission circuit **303** and a transmission coupler **304**, and the

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wireless communication apparatus **302** includes a reception circuit **306** and a reception coupler **305**. Wireless communication in the wireless communication system **300** is performed by transmitting a signal generated in the transmission circuit **303** from the transmission coupler **304**, receiving the signal by the reception coupler **305**, and generating a signal waveform at the reception circuit **306**. The transmission coupler **304** includes a transmission coupler **307** and a transmission coupler **308**.

While the transmission coupler **304** includes these two transmission couplers **307** and **308**, three or more transmission couplers may be included.

The transmission coupler **307** includes a signal line **309**, a distributor **310**, a signal line **311**, a coaxial cable **312**, and a terminating resistor **313** on the signal line **311**. The signal line **309** is connected to the distributor **310**, and the distributor **310** is connected to one end of the signal line **311** and one end of the coaxial cable **312**. The other end of the signal line **311** is terminated at the terminating resistor **313** at an end of the transmission coupler **307**.

The transmission coupler **308** includes a signal line **314** and a terminating resistor **315** on the signal line **314**. The other end of the coaxial cable **312** connected to the distributor **310** in the transmission coupler **307** is connected to one end of the signal line **314** in the transmission coupler **308**. The other end of the signal line **311** and the one end of the signal line **314** face each other.

The signal generated in the transmission circuit **303** is transmitted to the transmission coupler **307** and is distributed by the distributor **310**. One distributed signal travels on the signal line **311**, and is terminated at the terminating resistor **313**. The other distributed signal travels on the coaxial cable **312** and the signal line **314**, and is terminated at the terminating resistor **315**. The length and the type of the coaxial cable **312** are changed such that the phases of the one distributed signal and the other distributed signal are continuous at the end of the signal line **314** facing the signal line **311** and the end of the signal line **311** facing the signal line **314**. The length of the signal line **311** may be changed instead.

FIG. 4 illustrates a stereoscopic configuration example of a communication coupler in a concrete manner. The transmission coupler **304** and the reception coupler **305** are arranged to face each other in close proximity. If the reception coupler **305** travels in the longitudinal direction of the transmission coupler **304**, it is desirable that the transmission coupler **307** and the transmission coupler **308** are arranged substantially in the longitudinal direction.

In the present embodiment, single-ended signal communication is performed, but a differential signal can be used. Differential signal communication is performed with a transmission coupler and a reception coupler each provided with two signal lines. In the present embodiment, a communication coupler provided with a terminating resistor is a transmission coupler, and a communication coupler without a terminating resistor is a reception coupler, but the combination of a communication coupler without a terminating resistor as a transmission coupler and a communication coupler with a terminating resistor as a reception coupler may be used.

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The configuration according to the present embodiment of the present disclosure reduces splits at rising and falling edges of a reception signal waveform, providing high-speed wireless communications.

While the present disclosure has been described with reference to embodiments, it is to be understood that the disclosure is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of priority from Japanese Patent Application No. 2022-081887, filed May 18, 2022, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. A wireless communication apparatus comprising:

a communication coupler including a plurality of substrates including a first substrate and a second substrate, and a connection signal line; and

a circuit configured to transmit or receive a signal to or from the communication coupler,

wherein the first substrate and the second substrate are opposed to each other in a longitudinal direction of the communication coupler,

wherein the first substrate includes a first signal line, a second signal line, a third signal line, a distributor, and a terminating resistor,

wherein at the distributor, the first signal line branches into the second signal line and the third signal line, wherein the second signal line is terminated at the terminating resistor at an end of the first substrate,

wherein the second substrate includes a fourth signal line, wherein the connection signal line connects the third signal line and the fourth signal line, and

wherein a facing portion of the second signal line and a facing portion of the fourth signal line are opposed to each other, and at least one of the third signal line or the connection signal line is configured such that a phase of a signal at the facing portion of the second signal line and a phase of a signal at the facing portion of the fourth signal line are continuous.

2. The wireless communication apparatus according to claim 1,

wherein the communication coupler is a transmission coupler configured to transmit a signal to a reception coupler included in another communication apparatus, and

wherein the circuit generates the signal, and transmits the generated signal to the transmission coupler.

3. The wireless communication apparatus according to claim 1,

wherein the communication coupler is a reception coupler configured to receive a signal from a transmission coupler included in another communication apparatus, and

wherein the circuit forms a signal waveform based on the signal received by the reception coupler.

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